

Mathematics A

Unit 2 • Chapter OL-T33 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

3 classified items • 10 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T33 • Expertise Q16+ • 3 items • 10 marks

Chapter	Items	Marks	Starts
01. OL-T33 Graphs of Functions	3	10	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Graphs of Functions

Unit 2 • Expertise • 10 marks • 3 item(s)

November 2024 | 4MA1-1H Q23 | 5 marks

Infer a graph parameter from two intersections

January 2022 | 4MA1-2H Q21 | 2 marks

Estimate roots from axis crossings

January 2022 | 4MA1-2H Q21 | 3 marks

Draw a line to solve a nonlinear equation

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q23 M01

5 marks

- 23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.
- C** and **L** intersect at the points A and B
- The coordinates of point A are (p, k)
The coordinates of point B are (q, k)
- Given that $p - q = 14$
- find the value of k
Show clear algebraic working.

4MA1-1H Q23 M01

5 marks

$$k = \dots\dots\dots$$

WORKING SPACE

Use the chord separation to find k**5 marks****M01 – Use the chord separation to find k****Use the equal y-values**

$$p^2 - 8p - 9 = k \text{ and } q^2 - 8q - 9 = k.$$

Use symmetry

The axis is $x=4$, so $(p+q)/2=4$; with $p-q=14$, $p=11$, $q=-3$.

Substitute

$$k = 11^2 - 8(11) - 9 = 24.$$

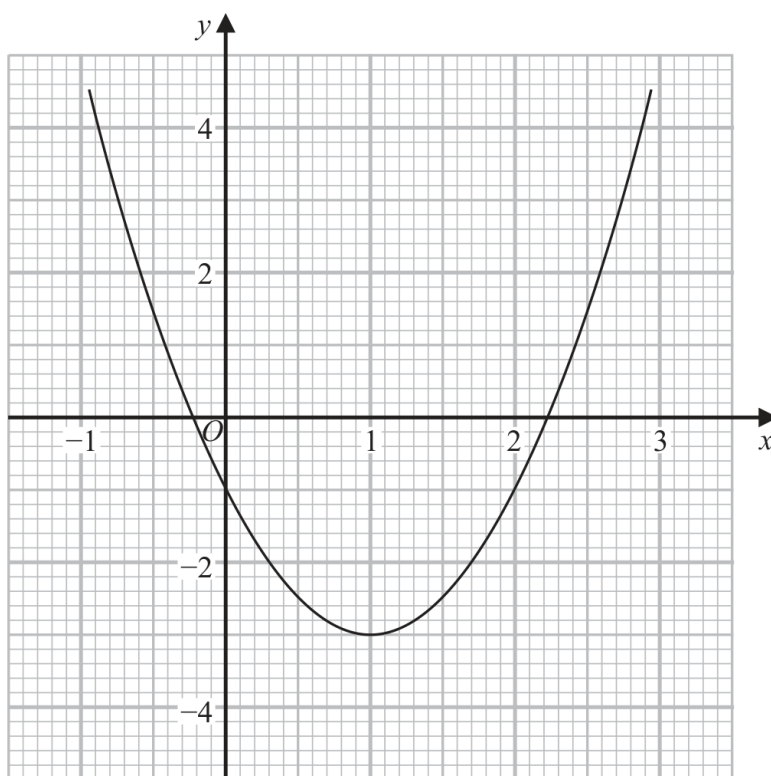
Final answer

$$k = 24$$

4MA1-2H Q21 Q21(a)

2 marks

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



- (a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$.
Give your solutions correct to one decimal place.

.....
(2)

Continue your method

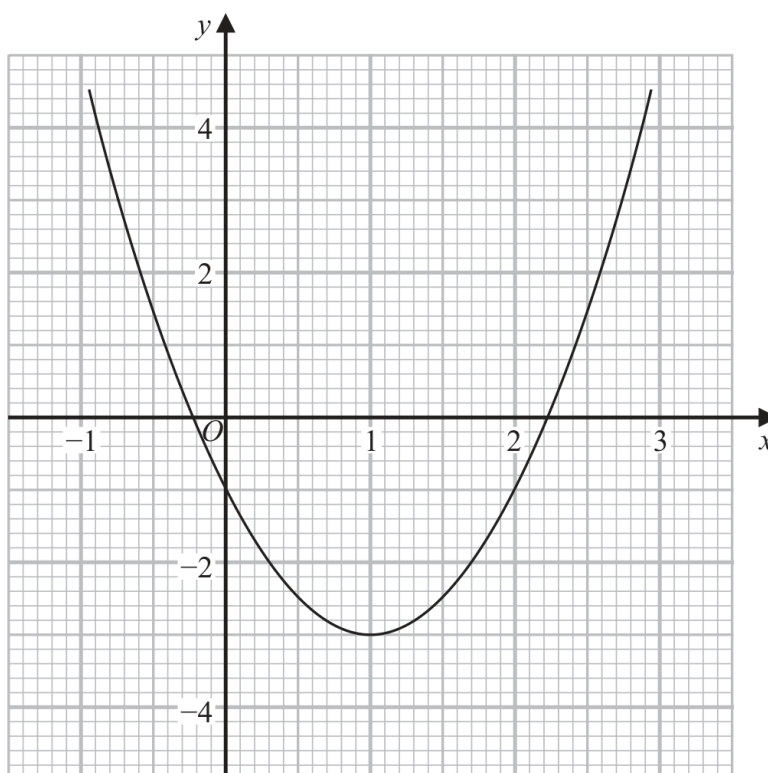
2 marks

WORKING SPACE

4MA1-2H Q21 Q21(a)

2 marks

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



- (a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$.
Give your solutions correct to one decimal place.

.....
(2)

Chapter: Graphs of Functions

2 marks

Q21(a) – Chapter: Graphs of Functions

Family: Solve using graph roots or intersections • Variant: Estimate roots from axis crossings

Method

The x-intercepts of $y=2x^2-4x-1$ are approximately $x=-0.2$ and $x=2.2$.

Why: Read where the curve crosses the x-axis.

Method

Both values are given to one decimal place.

Why: Round the graph readings as requested.

Final answer

roots: -0.2 and 2.2

4MA1-2H Q21 Q21(b)

3 marks

- (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

.....
(3)

4MA1-2H Q21 Q21(b)

3 marks

- (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

.....
(3)

Chapter: Graphs of Functions

3 marks

Q21(b) – Chapter: Graphs of Functions

Family: Solve using graph roots or intersections • Variant: Draw a line to solve a nonlinear equation

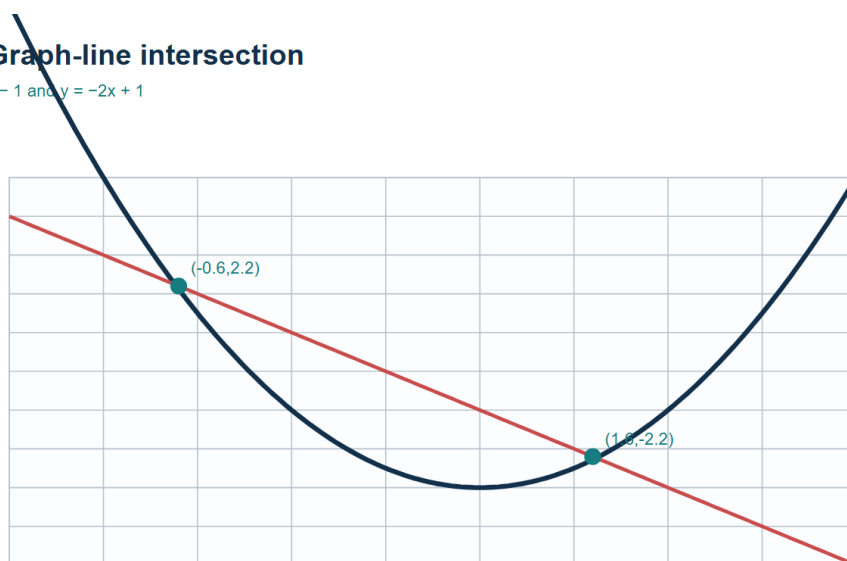
MethodFor $y=x^2-x-1$, use $y=-2x+1$ as the straight line.*Why: Rearrange $x^2-x-1=0$ to $x^2-x=1$ and compare with the displayed quadratic.***Method**Draw $y=-2x+1$ through $(0,1)$ and $(-1,3)$.*Why: Use two points to draw the line on the supplied grid.***Method**The intersections give $x \approx -0.6$ and $x \approx 1.6$.*Why: Read the two x -coordinates to one decimal place.***Final answer**roots: -0.6 and 1.6

Chapter: Graphs of Functions

3 marks

Q21 • Graph-line intersection

$$y = 2x^2 - 4x - 1 \text{ and } y = -2x + 1$$

x-intersections: -0.6 and 1.6

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